

Faculty of Computer Applications and IT

Subject: Embedded Systems (230701302)

Chapter – 3 : Embedded Hardware Building Blocks and the Embedded Board

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- This material is prepared for the students of MCA, FCAIT – PG Programme, GLS University considering the syllabus for the subject “Embedded Systems”
- The presentation included here outlines only the main points and key highlights of the course. It does not cover the entire syllabus in detail. For comprehensive information and complete understanding of the course material, please refer to the full syllabus and associated readings.

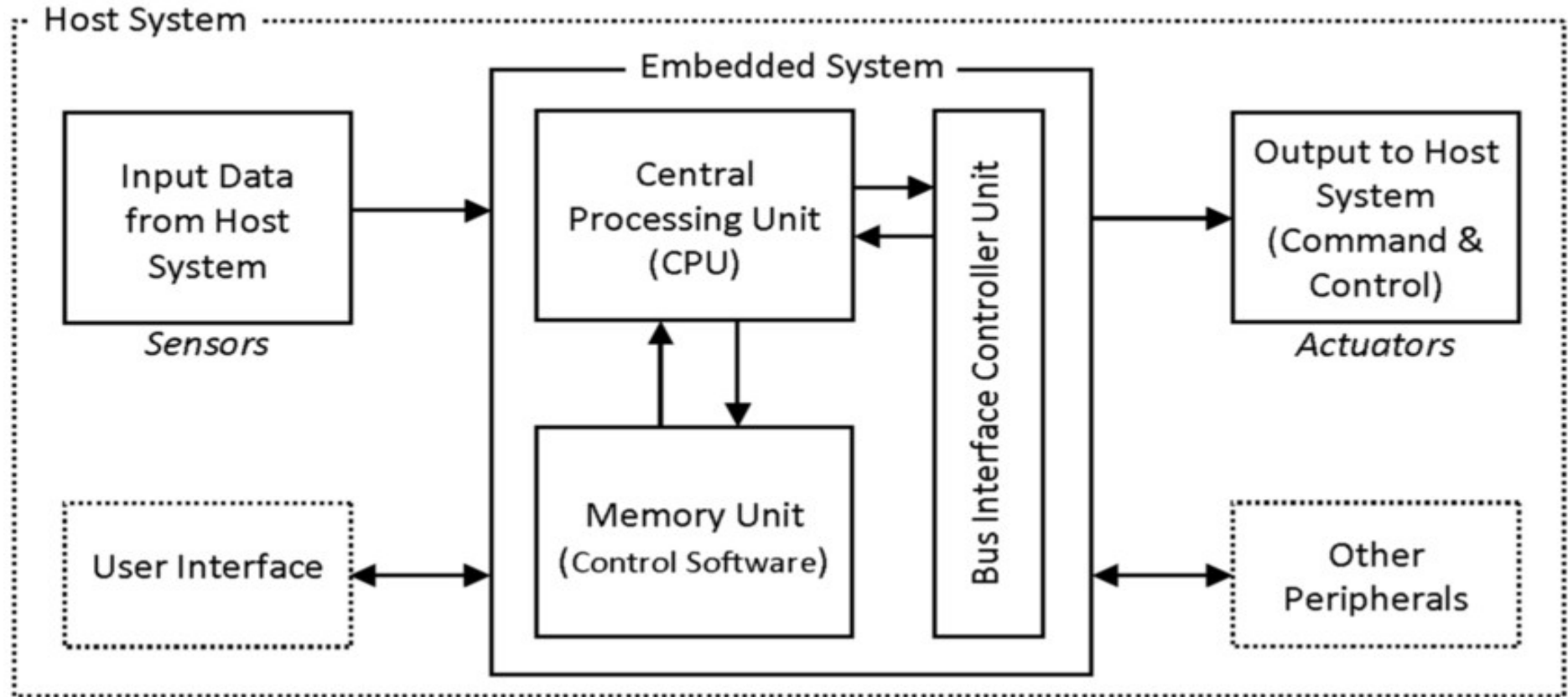
Hardware Drawings

- Different types of engineering hardware drawings
 - Block diagrams
 - Schematics
 - Wiring diagrams
 - Logic diagrams/prints
 - Timing diagrams

Block Diagrams

- A basic overview of the hardware, with implementation details abstracted out
- Reflect the actual physical layout of a board containing the major components
- The symbols used within a block diagram are simple, such as squares or rectangles for chips, and straight lines for buses
- Useful in communicating a basic overview of the hardware

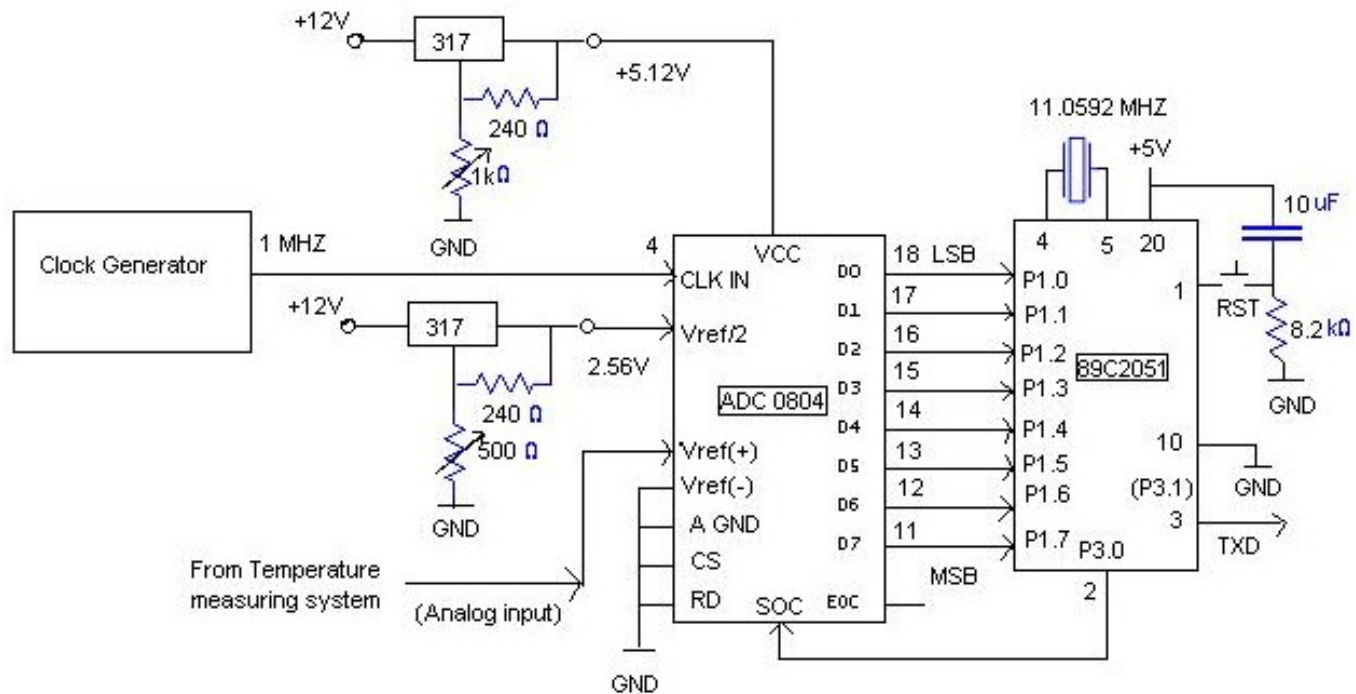
Block Diagrams



Schematics

- Electronic circuit diagrams
- More detailed view of all of the devices within a circuit
- Everything from processors down to resistors
- Not meant to depict the physical layout of the board or component
- Provides information on the flow of data in the system
- Schematic symbols are used to depict all of the components within the system

Schematics



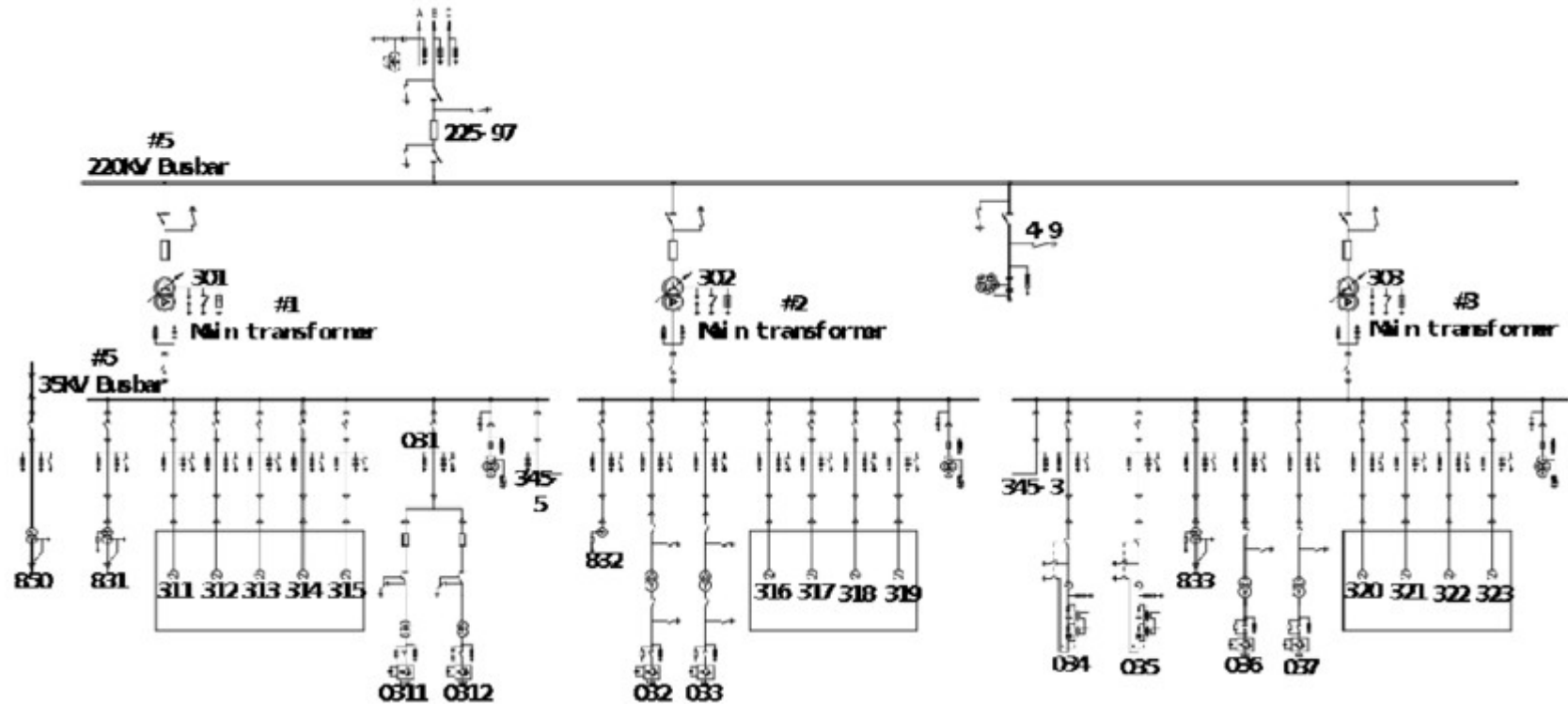
Schematics

- Some of the conventions and rules include
 - Title section
 - Schematic symbols with label
 - Abbreviations and prefixes
 - Functional groups
 - I/O and Voltage Source/Ground Terminals

Wiring diagrams

- Represent the bus connections between the major and minor components on a board or within a chip
- Vertical and horizontal lines are used to represent the lines of a bus

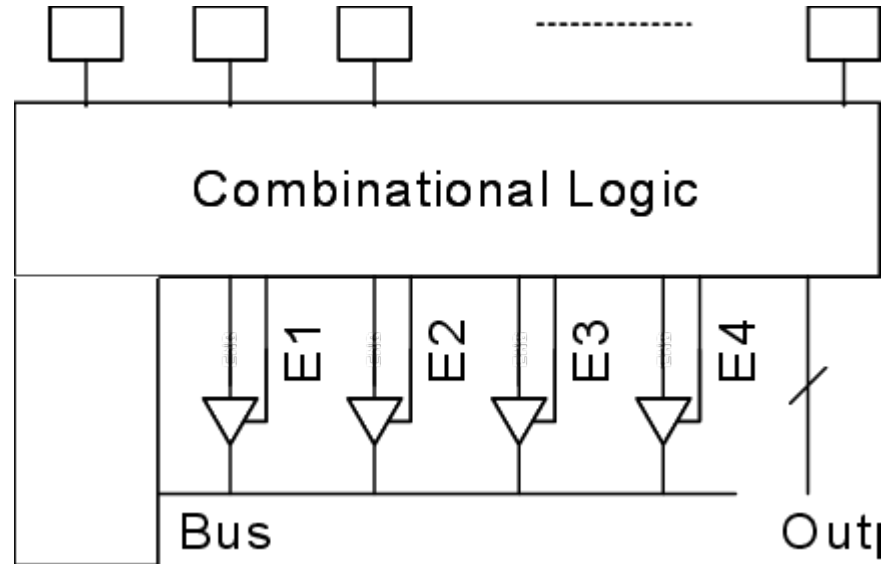
Wiring diagrams



Logic diagrams/prints

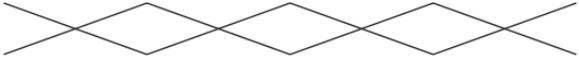
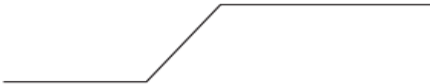
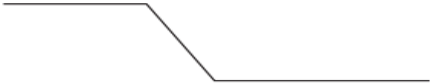
- Used to show a wide variety of circuit information using logical symbols and logical inputs and outputs
- Useful in simplifying certain types of circuits in order to understand how they function
-

Wiring diagrams



Timing diagram

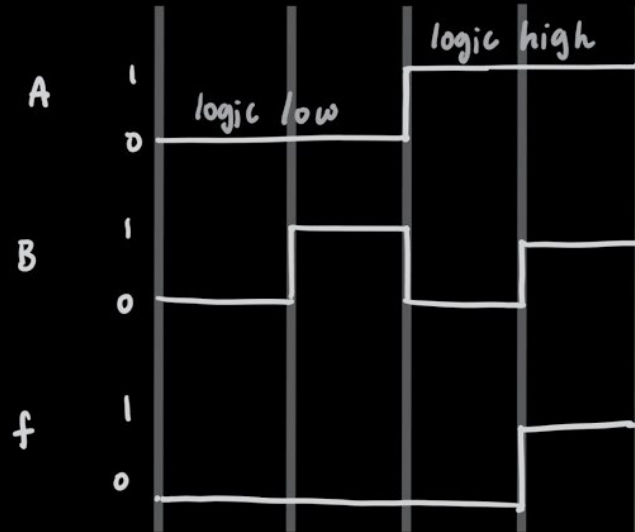
- Display timing graphs of various input and output signals of a circuit
- It also shows the relationships between the various signals

Symbol	Input Signals	Output Signals
	Input signal must be valid	Output signal will be valid
	Input signal doesn't affect system, will work regardless	Indeterminate output signal
	Garbage signal (nonsense)	Output signal not driven (floating), tristate, HiZ, high impedance
	If the input signal rises	Output signal will rise
	If the input signal falls	Output signal will fall

Timing diagram

Digital Logic

Timing Diagrams



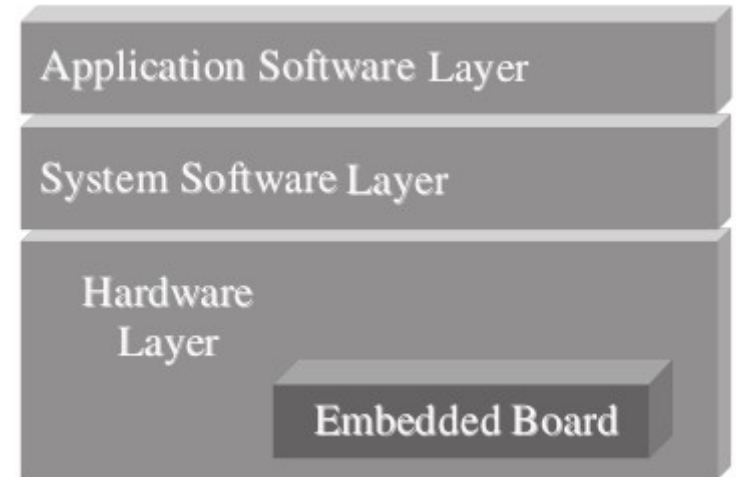
A	B	f
0	0	0
0	1	0
1	0	0
1	1	1

Traister and Lisk method

- **Step 1:** Learning the basic symbols that can make up the type of diagram, such as timing or schematic symbols. To aid in the learning of these symbols, rotate between this step and steps 2 and/or 3.
- **Step 2:** Reading as many diagrams as possible, until reading them becomes boring (in that case rotate between this step and steps 1 and/or 3) or comfortable (so there is no longer the need to look up every other symbol while reading).
- **Step 3:** Writing a diagram to practice simulating what has been read, again until it either becomes boring (which means rotating back through steps 1 and/or 2) or comfortable.

The Embedded Board and the von Neumann Model

- In embedded devices, all the electronics hardware resides on a board
 - Printed Wiring board (PW)
 - Printed Circuit Board (PCB)
 - thin sheets of fiberglass



Embedded Board

- Major hardware components of most boards based upon the major elements defined by the von Neumann model:
 - Central Processing Unit (CPU) – the master processor
 - Memory – where the system's software is stored
 - Input Device(s) – input slave processors and relative electrical components
 - Output Device(s) – output slave processors and relative electrical components
 - Data Pathway(s)/Bus(es) – interconnects the other components

Powering the Hardware

- Rate that energy is expended or work is performed
- Accurate power and energy calculations must be done for all elements on an embedded board to determine the power consumption requirements of that particular board
- ES uses both AC and DC voltage sources
- Some embedded boards integrate or plug into power supplies
- Power supplies can be either AC or DC
- AC-to-DC converter
- Embedded board require 3.3, 5, or 12 volts

- Batteries as voltage sources
 - limited life
 - either recharged or replaced

Analog vs. Digital Signals

- A digital system processes only digital data
- Two voltages represent “0” and “1”
- No voltage (0 volts) : ground, VSS, or low
- VCC, VDD or HIGH : 3, 5, or 12 volts
- “0” as low and “1” as high

Basic Hardware Materials

- All electronic devices used on a board or that come into contact with an embedded board are made up of materials that are generally classified as
 - Conductors
 - Insulators
 - Semiconductors
- Differ according to the ability of the materials to conduct an electric current

Conductors

- Materials that have fewer impediments to an electric current
 - more easily lose/gain valence electrons
 - they have three or fewer valence electrons
- Metals are conductors
- Three or fewer valence electrons



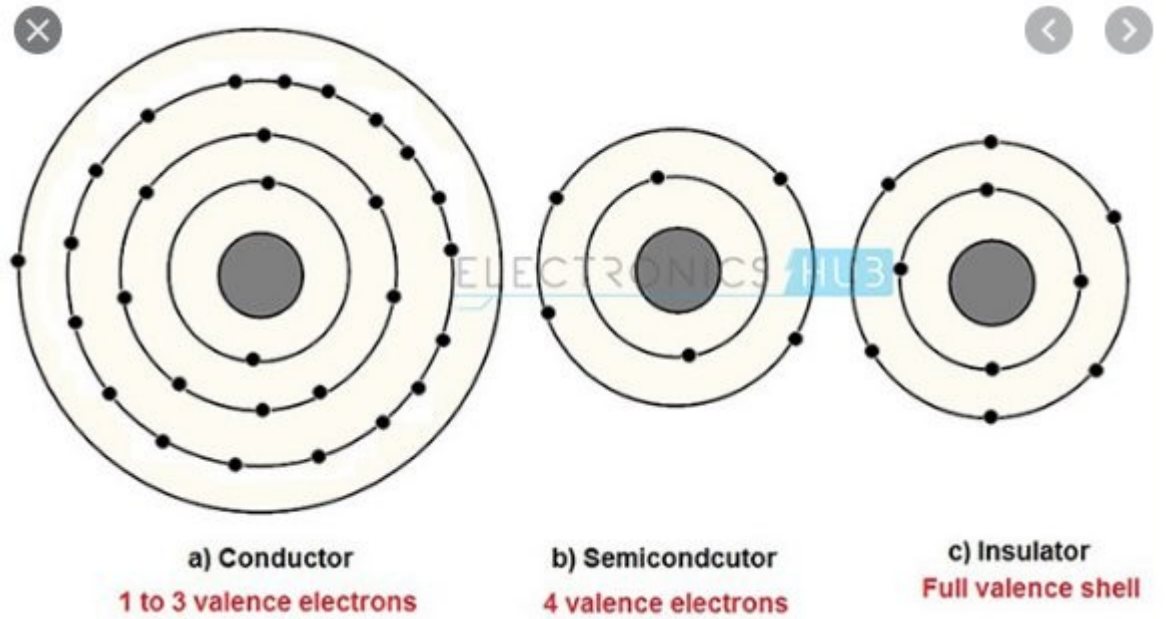
Insulators

- Five or more valence electrons
- Insulators are typically not the main materials used in buses
- Non-metals, such as air, paper, oil, plastic, glass, and rubber, are usually considered insulators

Semiconductors

- Usually have four valence electrons
- Classified as materials whose base elements have a conductive nature that can be altered by introducing other elements into their structure
- Semiconductive material has the ability to behave both as a conductor and as an insulator
- silicon and germanium

Comparision



Common Passive Components on Boards and in Chips

- Can be integrated to form processors and memory chips
- Part of the circuitry

Common Passive Components on Boards and in Chips

- Resistors
- Capacitors
- Inductors

The Resistor

- Devices made up of conductive materials that have had their conductivity altered in some fashion to allow for an increase in resistance
 - carbon-composition resistors
 - wire-wound and carbon-composition
- Used to control current and voltage
- Commonly used in a variety of circuitry both on boards and integrated into processor or memory chips

The Resistor

- Properties
 - Tolerance in %
 - Power rating
 - Reliability level rating in %
 - Temperature coefficient of resistance

Reference

- Tammy Noergaard (2005). Embedded Systems Architecture: A Comprehensive Guide for Engineers and Programmers. Elsevier